

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:22-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

22-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-24(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

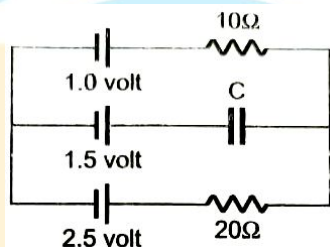
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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

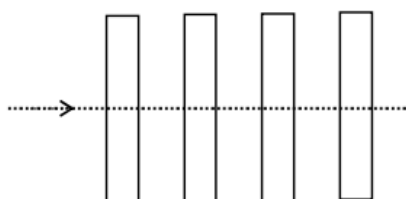
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

- An object of mass 5 kg falls from rest through a vertical distance of 20 m and reaches ground with a velocity of 10 m/s. The work done by air resistance ($g = 9.8 \text{ ms}^{-2}$) is:
A) +730 J B) -730 J C) +980 J D) -980 J
- A point source of light is placed 4 m below the surface of water of refractive index $5/3$. The minimum diameter of a disc, which should be placed over the source, on the surface of water to cut-off all light coming out of water is just greater than:
A) 8 m B) 4 m C) 3 m D) 6 m
- In the following circuit diagram the potential difference across the plates of the capacitor C in steady state is:



- A) 2.5 volt B) 1.5 volt C) 1.0 volt D) zero
- If $\vec{E}(x, t) = E_0 \sin(\omega t - kx) \hat{j}$ and $\vec{B}(x, t) = B_0 \sin(\omega t - kx) \hat{k}$ are the electric field and magnetic field vectors respectively in an electromagnetic wave. What would be the direction of propagation of the EM Wave?
A) $\hat{i}$ B) $\hat{i} + \hat{j}$ C) $\hat{k}$ D) $\hat{j} + \hat{k}$
- A beam of unpolarized light of intensity I_0 falls on a system of four identical linear polarizers placed in a line as shown in the figure. The transmission axis of any two successive polarizers make an angle of 30° with each other. If the transmitted light has intensity I , the ratio $\frac{I}{I_0}$ is

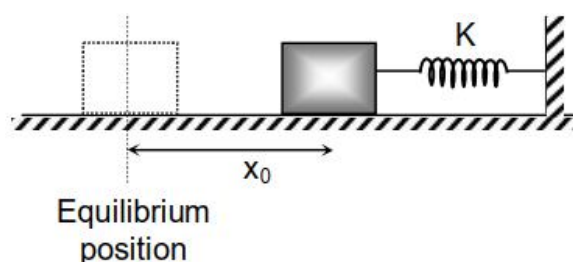


- A) $\frac{81}{256}$ B) $\frac{9}{16}$ C) $\frac{27}{64}$ D) $\frac{27}{128}$

6. For an ideal gas, pressure P varies with volume V as $P \propto V^{-2/3}$. Then the bulk modulus B of the gas during this process will be proportional to
- A) $V^{-2/3}$ B) $V^{-5/3}$ C) V^{-2} D) $V^{-\frac{2}{5}}$
7. Three infinitely long wires, each carrying equal current in the same direction are placed in the xy -plane along $x = 0, +d$ and $-d$. On the xy -plane, the magnetic field vanishes at
- A) $x = \pm \frac{d}{2}$ B) $x = \pm d \left(1 + \frac{1}{\sqrt{3}} \right)$
- C) $x = \pm d \left(1 - \frac{1}{\sqrt{3}} \right)$ D) $x = \pm \frac{d}{\sqrt{3}}$
8. Match the following two columns for L-C-R series AC circuit.

	Column I		Column II
A.	At resonance frequency	(p)	Power factor=0
B.	No resistance in the circuit	(q)	Power factor=1
C.	Only resistance in the circuit	(r)	Circuit is capacitive
D.	Frequency greater than the resonance frequency	(s)	Circuit is inductive

- A) A-q, B-p, C-q, D-s B) A-p, B-p, C-q, D-s
- C) A-s, B-p, C-q, D-p D) A-q, B-p, C-s, D-r
9. Initially spring is compressed by distance x_0 from equilibrium position. At this compression block is given velocity $\sqrt{\frac{3K}{m}}x_0$, so that compression in the spring starts increasing and block starts S.H.M. (Spring constant is K). Equation of motion of the block is: (Take origin as equilibrium position and take positive direction of x -axis towards right)



A) $x = 3x_0 \sin\left(\sqrt{\frac{K}{m}}t + \frac{\pi}{3}\right)$

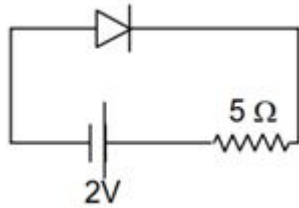
B) $x = 2x_0 \sin\left(\sqrt{\frac{K}{m}}t + \frac{\pi}{6}\right)$

C) $x = 3x_0 \sin\left(\sqrt{\frac{K}{m}}t + \frac{\pi}{6}\right)$

D) $x = 2x_0 \sin\left(\sqrt{\frac{K}{m}}t + \frac{\pi}{3}\right)$

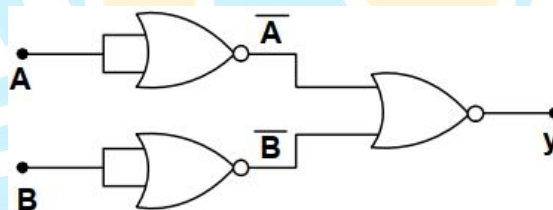
10. Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

Assertion (A): No appreciable current flows in the given circuit.

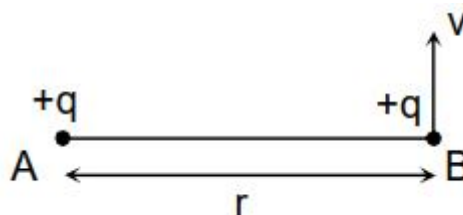


Reason (R): In reverse biased diode offers infinite resistance.

- A) Both A and R are true and R is the correct explanation of A
 B) Both A and R are true but R is not the correct explanation of A
 C) A is true but R is false
 D) A is false and R is also false
11. Identify the logic operation performed by the circuit given below

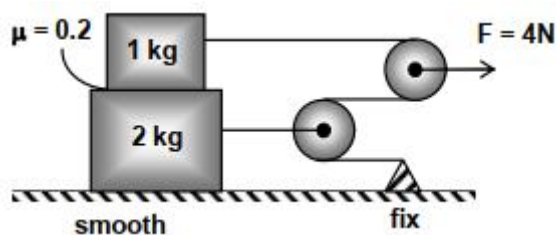


- A) NOT B) AND C) OR D) NAND
12. In the figure shown, A is a fixed charge. B (of mass m) is given a velocity V perpendicular to line AB. At this moment the radius of curvature of the resultant path of B is



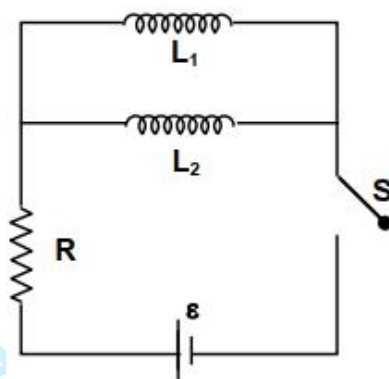
- A) 0 B) $\frac{r}{2}$ C) $\frac{4\pi\epsilon_0 r^2 m v^2}{q^2}$ D) r

13. All the pulleys and strings are ideal and massless and the system shown is initially at rest. Now a force $F = 4\text{ N}$ is applied on the pulley shown. Then the frictional force acting between the blocks is (take $g = 9.8 \text{ m/s}^2$)



- A) 2 N B) 1.96 N C) 3 N D) 0 N
14. A small smooth metal sphere of radius r and density ρ falls from rest in a viscous liquid of density σ and coefficient of viscosity η . Due to viscous force, heat is produced. The rate of production of heat when the sphere has acquired the terminal velocity is proportional to
- A) r B) r^3 C) r^4 D) r^5
15. A parallel beam of light having intensity $I = 10 \text{ Wm}^{-2}$ is continuously falling on emitter plate in the photoelectric effect setup in saturated state at angle of 37° with the normal. If area of plate is 1 m^2 and wavelength of light is 621 nm and one photon is capable to eject one electron. Then saturation current in the circuit is ($\lambda_{\text{th}} > 621 \text{ nm}$) close to
- A) 3 A B) 4 A C) 1.8 A D) 3.2 A
16. A point mass performs straight line motion along positive x -axis. At $t = 0$ point mass is at point $A(x_1, 0)$. It moves such that its velocity is given by $v = \frac{a}{x}$, where a is positive constant and x is the x -coordinate of position vector of point mass at a certain time t . Find the time required to move from A to $B(x_2, 0)$
- A) $\frac{x_2^2 - x_1^2}{2a}$ B) $\frac{x_2^2 - x_1^2}{a}$ C) $\frac{2x_2^2 - x_1^2}{2a}$ D) $\frac{2x_2^2 - x_1^2}{a}$

17. In the circuit shown, switch S is closed at $t = 0$. Then the ratio $\frac{U_1}{U_2}$ of potential energy stored in the inductors L_1 and L_2 in steady state is



- A) $\frac{L_1}{L_2}$ B) $\frac{L_2}{L_1}$ C) $\left(\frac{L_1}{L_2}\right)^2$ D) $\left(\frac{L_2}{L_1}\right)^2$
18. A rigid bar of mass M is supported symmetrically by three vertical wires each of length ℓ . Those at each end are of copper and the middle one at midpoint is of iron. The ratio of their diameters $\frac{d_{\text{copper}}}{d_{\text{iron}}}$, if each is to have the same tension, is equal to
- A) $\frac{Y_{\text{copper}}}{Y_{\text{iron}}}$ B) $\sqrt{\frac{Y_{\text{iron}}}{Y_{\text{copper}}}}$ C) $\frac{Y_{\text{iron}}^2}{2Y_{\text{copper}}^2}$ D) $\frac{2}{3} \frac{Y_{\text{iron}}}{Y_{\text{copper}}}$
19. Statement-I: The mass number of a nucleus is always less than its atomic number
Statement-II: Mass number of a nucleus may be equal to its atomic number.
- A) Both Statements are correct. B) Only Statement-I is correct.
C) Only Statement-II is correct D) Both Statements are incorrect
20. A vernier calipers has 1 mm marks on the main scale. It has 10 equal divisions on the vernier scale which match with 8 main scale divisions. What will be the least count of vernier calipers in mm.
- A) 0.1 B) 0.2 C) 0.3 D) 0.5

SECTION-II

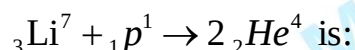
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

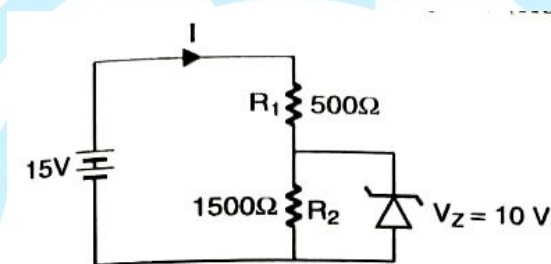
Marking scheme: +4 for correct answer, -1 in all other cases.

21. A body of mass 2 kg is floating in water with half its volume submerged. What would be the external vertical force required in static equilibrium to wholly submerge it into water? (in N) (take $g = 10 \text{ m/s}^2$)

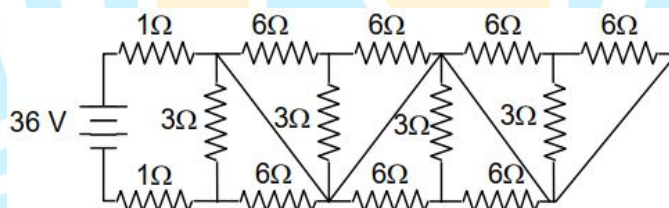
22. If binding energies per nucleon for ${}_3\text{Li}^7$ and ${}_2\text{He}^4$ are 5.60 MeV and 7.06 MeV respectively, then the energy released in the nuclear process: (in MeV) (Round-off to the nearest integer)



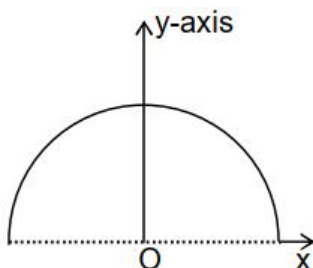
23. In the given circuit the current through the Zener diode is $\left(\frac{10}{n}\right) \text{ mA}$. Find the value of n



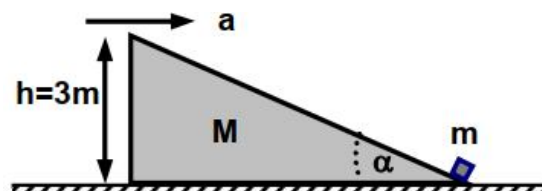
24. Find the current (in A) through battery in following circuit.



25. A semicircular disc of radius $R = \pi$ meter and mass M has surface mass density ρ in SI units which varies as $\rho = Cr^2$, where r is the distance (in m) from the center of the base O and C is a constant. The distance of center of mass of the semicircular disk from O is $\left(\frac{n}{10}\right) \text{ meter}$. Find the value of n .



26. A string of length 1m fixed at both ends is vibrating in 1st overtone. Tension in string is 200 N and linear mass density is 5 gm/m. Frequency (in Hz) of these vibrations is :
27. A ring of mass m is placed on a rough horizontal surface with its plane vertical. A horizontal impulse J in the plane of the ring is applied on the ring of mass m along a line passing through its centre. The linear velocity of the centre of the ring once it starts pure rolling is $\frac{J}{km}$. Find the value of k .
28. 3.2 kg of ice at -10°C just melts when steam of mass m , at 100°C is passed through it. Then find mass m (in grams)
(specific heat of ice = $0.5 \text{ cal/gm}^\circ\text{C}$, latent heat of ice = 80 cal/gm , latent heat of vaporization of water = 540 cal/gm , specific heat of water = $1 \text{ cal/gm}^\circ\text{C}$)
29. A soap bubble of radius R , surface tension T exists in equilibrium in vacuum. A monoatomic gas is trapped inside the soap bubble. Charge Q is given to bubble such that bubble expands from radius a to radius $2a$ under reversible adiabatic condition. If surface charge density on the bubble surface is found to be $\sqrt{\frac{x\epsilon_0 T}{ya}}$. Find the value of x/y . (x, y are positive co-primes)
30. A wedge of mass $M = 10\text{kg}$, height $h = 3\text{m}$ and angle of inclination $\alpha = 37^\circ$ is at rest at horizontal surface. There is a small point-like object (mass $m = 0.5\text{kg}$) next to the slope as shown in the figure. At what acceleration must wedge be moved in order that the point like object reaches its top in a time $t = 5\text{s}$ (Neglect the friction between point like object and wedge) ($g = 10 \text{ m/s}^2$)



SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. The decreasing order of the second ionization potential of K, Ca and Ba is

- A) $K > Ca > Ba$ B) $Ca > Ba > K$ C) $Ba > K > Ca$ D) $K > Ba > Ca$

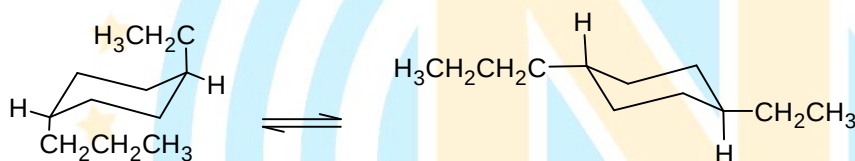
32. Consider the following solution

- (I) 1M Sucrose (II) 1M KCl (III) 1M Benzoic acid (IV) 1M $(NH_4)_3 PO_4$

Which of the following is/are true ?

- A) All solutions are isotonic B) III is hypertonic of II, I, IV
C) I, II and III are hypertonic of IV D) IV is hypertonic of I, II and III

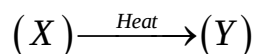
33. Consider the following equilibrium



Which of the following best describes the equilibrium constant K of this inter conversion?

- A) $K < 1$ B) $K > 1$ C) $K = 1$
D) K can not be deduced from this information.

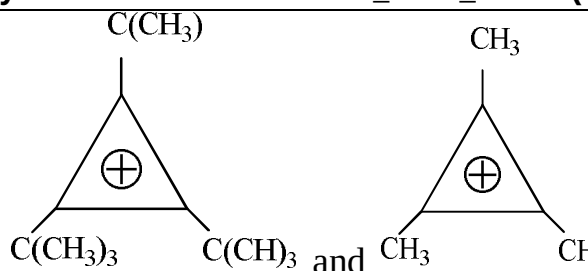
34. Compound (A) $\xrightarrow{\text{Conc } H_2SO_4}$ (X) Green oily liquid



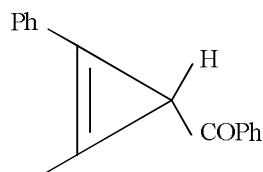
The (X) and (Y) are respectively:

- A) Mn_2O_7, MnO_2 B) Mn_2O_7, Mn_2O_3
C) $MnSO_4, Mn_2O_3$ D) Mn_2O_3, MnO_2

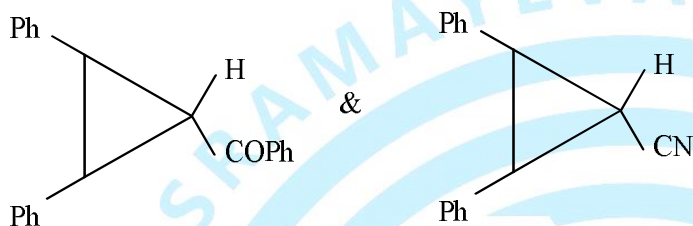
35. Which one of the following is correctly matched.
- A) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}$ is tetrahedral and $\text{Na}_2[\text{CoCl}_4]$ is square planar
- B) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}$ and $\text{Na}_2[\text{CoCl}_4]$ are tetrahedral
- C) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}$ is square planar and $\text{Na}_2[\text{CoCl}_4]$ is tetrahedral
- D) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}$ and $\text{Na}_2[\text{CoCl}_4]$ are square planar.
36. Which of the following will have two stereoisomeric forms?
- (I) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ (II) $\text{K}_3[\text{Fe}(\text{ox})_3]$
- (III) $[\text{CoCl}_2(\text{en})_2]^+$ (IV) $[\text{Co}(\text{Br})(\text{Cl})(\text{ox})_2]^{-3}$
- A) I only B) I and II only C) III and IV only D) I, II, III, IV
37. Which of the following statements are not correct?
- (I) All C – O bonds in CO_3^{2-} are equal but not in CO_2
- (II) All C – O bond lengths are same in acetic acid.
- (III) P – O bond order in PO_4^{3-} is 1.25
- (IV) The average formal charge on each oxygen atom in ClO_4^- is -0.25.
- A) I, II, III, IV B) I and II only C) III and IV only D) I, II, III only
38. Incorrect statement about SOF_4 is
- A) Hybrid orbital of central atom S has 60% P-character.
- B) Oxygen atom is present at equatorial position
- C) Angle between O–S–F can have value more than 90° .
- D) Sulphur atom shows complete octet.
39. An organic compound is treated with NaNO_2 and HCl under ice cold conditions and resulting solution is treated with 2-naphthol solution. a brilliant red colour indicates that the organic compound contains
- A) aliphatic – NH_2 group B) aromatic – NH_2 group
- C) $-\text{CONH}_2$ group D) $-\text{NO}_2$ group



40. Statement I: $\text{C}(\text{CH}_3)_3$ $\text{C}(\text{CH}_3)_3$ and CH_3 CH_3 are stable.



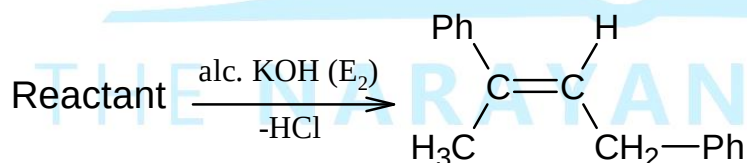
Statement II: Ph lose proton more slowly than



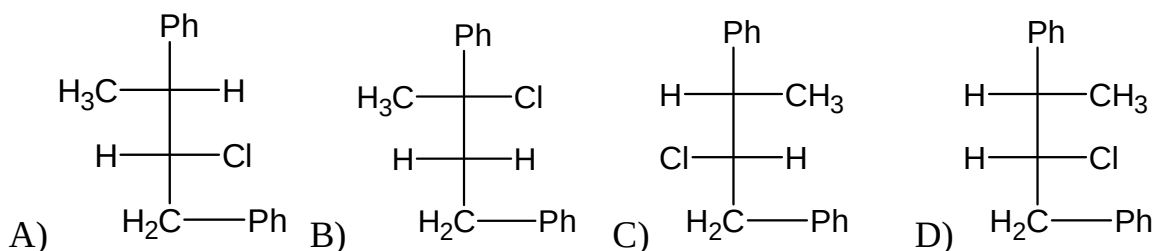
In the light of the above statements. Choose the most appropriate answer from the options given below.

- A) Both statement I and statement II are correct.
- B) Statement I is correct but statement II is incorrect.
- C) Statement I is incorrect but statement II is correct.
- D) Both statement I and statement II are incorrect.

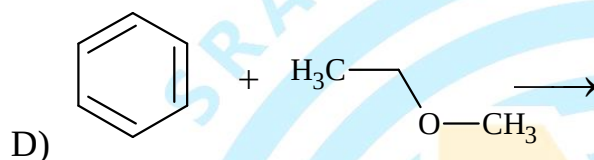
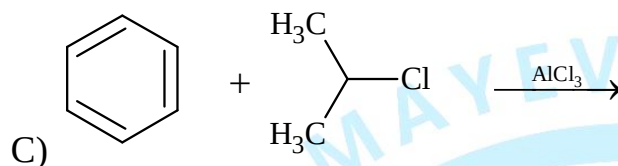
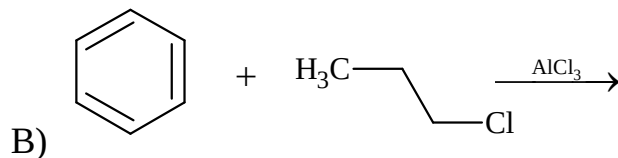
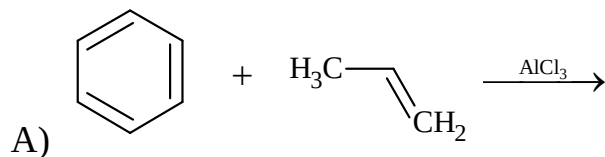
41.



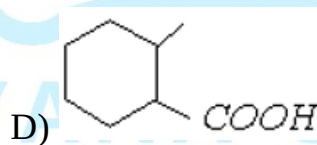
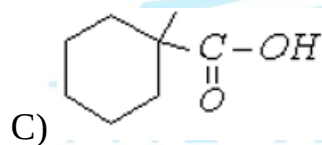
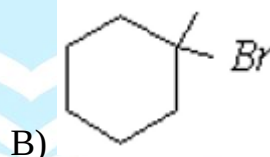
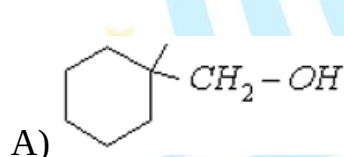
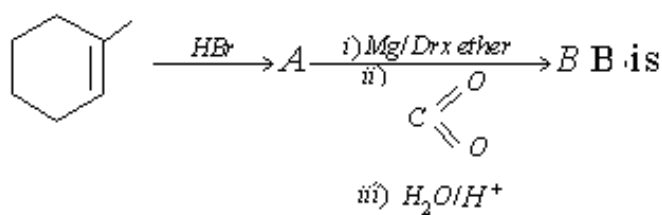
Which of the following molecule is not suitable reactant?



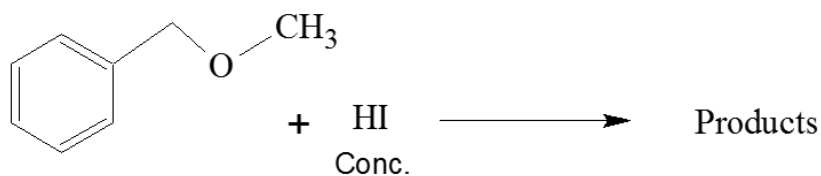
42. Isopropylbenzene can not be obtained by



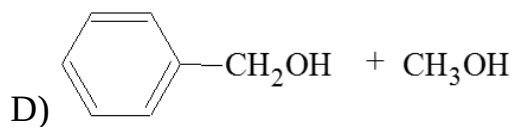
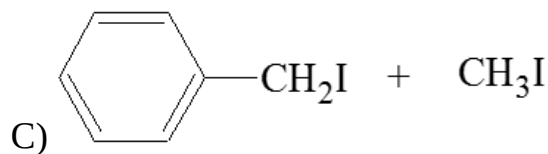
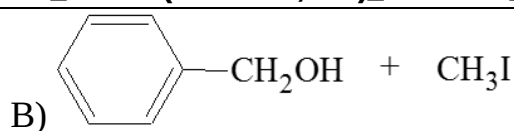
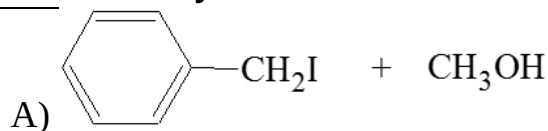
43.



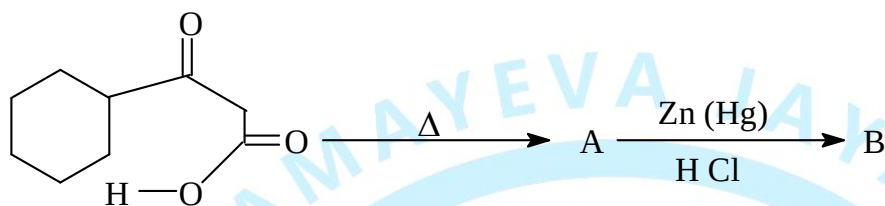
44.



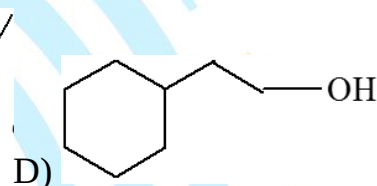
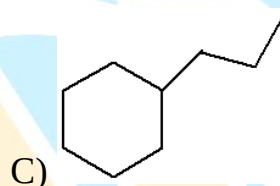
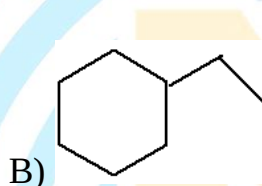
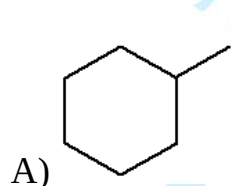
in above reaction products are:



45.



In the above reaction product B is



46. The correct sequence regarding the activity of the given acid derivative is



47. Which of the following statements are correct ?

(I) 1°, 2°, and 3° nitro compounds can be distinguished by HNO₂.

(II) 1° nitro compound with HNO₂ gives nitrolic acid, which gives blood-red colour with base.

(III) 2° nitro compound with HNO₂ gives pseudo nitrole, which gives blue colour with base.

(IV) 3° nitro compound does not react with HNO₂.

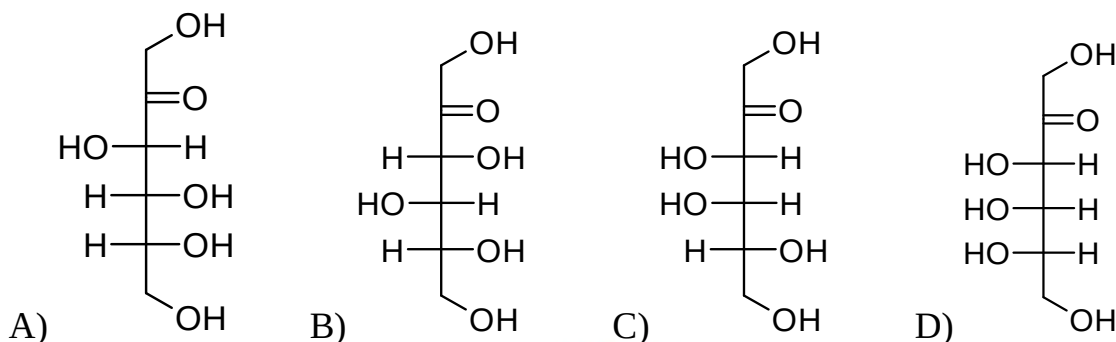
A) I, II only

B) II, III, IV only

C) I, II, III, IV

D) I, II, IV only

48. D-tagatose is a 2-ketobexose which forms same osazone as D-galactose. What is the structure of D-tagatose



49. Oxidation state of 'V' in $Rb_4Na[HV_{10}O_{28}]$ is

A) +5 B) +6 C) $+\frac{7}{5}$ D) +4

50. The value of $(n_2 + n_1)$ and $(n_2^2 - n_1^2)$ for He^+ ion in atomic spectrum are 4 and 8 respectively. The wavelength of emitted photon when electron jump from n_2 to n_1 is

A) $\frac{32}{9}R_H$ B) $\frac{9}{32}R_H$ C) $\frac{9}{32R_H}$ D) $\frac{32}{9R_H}$

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

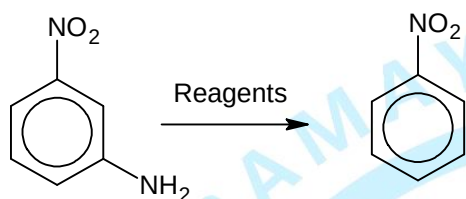
Marking scheme: +4 for correct answer, -1 in all other cases.

51. When SO_2 is passed into acidified $K_2Cr_2O_7$ solution, the oxidation state of sulphur in the product is ____
52. The total number of stereo isomers of the complex $[Rh(NO_2)(SCN)(en)_2]^+$ is ____
53. Based on VSEPR theory, the number of $90^\circ Cl - P - Cl$ angles in PCl_5 is X and the number of $120^\circ Cl - P - Cl$ angles in PCl_5 is Y then $X - Y$ is ____.
54. The monochloronitiated products can be obtained from all the isomeric alkanes having formula C_5H_{10} is xy . Sum of $x + y$ is

55. How many of the following are correct?

- a) Acetone can be prepared by Rosenmund's reduction
- b) The ratio of sp^2 and sp^3 hybridized carbons in phorone is 5 : 4
- c) Acetaldehyde reacts with Grignard's reagent but not acetone
- d) Mesitylene is the condensation product of acetone

56. How many of the following reagents are correct for the given reaction?



- a) (i) $\text{NaNO}_2 + \text{HCl}$, $0-5^\circ\text{C}$ (ii) H_3PO_2
- b) (i) $\text{KNO}_2 + \text{HBr}$, $0-5^\circ\text{C}$ (ii) Na_2SnO_2
- c) (i) HNO_2 (ii) $\text{C}_2\text{H}_5\text{OH}$ and heat
- d) (i) $\text{KNO}_2 + \text{HCl}$ (ii) $\text{H}_2\text{O}(\text{steam})$

57. The number of moles of periodic acid required to oxidize one mole of methyl glycoside is

58. Bond enthalpies of AB, A_2 and B_2 respectively are in the ratio 1 : 1 : 0.5. If ΔH for the formation of AB is -200 kJ , bond enthalpy of A_2 will be $x \times 10^2 \text{ kJ mole}^{-1}$. Find 'x' value.

59. How many mole of NH_4Cl must be added to one litre of 1.0 M NH_4OH to have a buffer of $\text{pH} = 9$. p^{K_b} of $\text{NH}_4\text{OH} = 5$

60. A solution containing Na_2CO_3 and NaOH requires 300 ml of 0.1N HCl using phenolphthalein as an indicator. Methyl orange is then added to the above titrated solution when a further 25 ml of 0.2N HCl is required. The amount of NaOH present in solution is ($\text{NaOH} = 40$, $\text{Na}_2\text{CO}_3 = 106$)

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. If $Z = x + iy$ be a complex number, $Z^{\frac{1}{3}} = a - ib$ and $\frac{x}{a} - \frac{y}{b} = \lambda(a^2 - b^2)$, (where x, y, a, b and λ are positive real numbers) then the value of λ is
- A) 2 B) 3 C) 4 D) 1
62. Statement 1: If $A = [a_{ij}]_{4 \times 4}$ such that $a_{ij} = \begin{cases} 2, & \text{when } i = j \\ 0, & \text{when } i \neq j \end{cases}$ then $\left\{ \frac{\det(\text{adj}(\text{adj } A))}{7} \right\} = \frac{1}{7}$
(where $\{\cdot\}$ denotes fractional part function)
- Statement 2: If A is matrix then $|\text{adj}(kA)| = k \cdot |A|$ (k is scalar)
- A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct Explanation for Statement-1
- B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- C) Statement-1 is True, Statement-2 is False
- D) Statement-1 is False, Statement-2 is True
63. If the roots of the equation $ax^2 + bx + c = 0$ be α and β and those of the equation $Ax^2 + Bx + C = 0$ be $\alpha + k$ and $\beta + k$ (where a, b, c, A, B, C and k are positive real numbers) then $\frac{b^2 - 4ac}{B^2 - 4AC}$ is
- A) $\frac{a^2}{A^2}$ B) 1 C) $\frac{a}{A}$ D) $\frac{A}{a}$
64. The mean and variance of 10 observations are found to be 10 and 5 respectively. On rechecking it is found that an observation 5 is incorrect. If the incorrect observation is replaced by 15, then the correct variance is
- A) 7 B) 8 C) 9 D) 4

65. If the probability of A to pass in an examination is 0.3 and for B is 0.5. Then the probability that either A or B pass in the examination is

- A) 0.65 B) 0.35 C) 0.8 D) 0.2

66. A random variable X has the distribution

X	2	3	4
$P(X = x)$	0.3	0.4	k

(where k is a positive real number)

Then, match the List-I to List-II

List – I

List – II

A) k

P) 0.6

B) mean

Q) 0.4

C) variance

R) 3

D) $P(2 < x < 4)$

S) 0.3

A) A – S, B – R, C – P, D – Q

B) A – Q, B – R, C – P, D – S

C) A – R, B – Q, C – P, D – S

D) A – P, B – Q, C – R, D – S

67. The number of irrational terms in the expansion of $\left(5^{\frac{1}{3}} + 2^{\frac{1}{4}}\right)^{100}$ is

- A) 94 B) 92 C) 93 D) 91

68. If $f''(x) > 0$ for all $x \in \mathbb{R}$, then the function g(x) given by, $g(x) = 2f\left(\frac{x^2}{2}\right) + f(6 - x^2)$

for all $x \in \mathbb{R}$, is decreasing for $x \in$

- A) $(2, \infty)$ B) $(-2, 0)$
C) $(-2, 0) \cup (2, \infty)$ D) $(-\infty, -2) \cup (0, 2)$

69. If $\hat{a}, \hat{b}, \hat{c}$ are unit vectors then $|\hat{a} - \hat{b}|^2 + |\hat{b} - \hat{c}|^2 + |\hat{c} - \hat{a}|^2$ does not exceed
- A) 6 B) 7 C) 8 D) 9
70. Statement 1: The shortest distance between the lines $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ is 9
- Statement 2: Lines $\vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$ and $\vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k})$, where $\lambda, \mu \in \mathbb{R}$ are coplanar.
- A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct Explanation for Statement-1
- B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- C) Statement-1 is True, Statement-2 is False
- D) Statement-1 is False, Statement-2 is True
71. The locus of the image of the point (2, 3) with respect to the line $(x - 2y + 3) + \lambda(2x - 3y + 4) = 0$ ($\lambda \in \mathbb{R}$)
- A) $x^2 + y^2 - 2x - 4y + 4 = 0$ B) $x^2 + y^2 + 2x - 4y + 4 = 0$
- C) $x^2 + y^2 - 3x - 4y - 4 = 0$ D) $x^2 + y^2 - 2x - 4y + 3 = 0$
72. Let A(1, 2), B(3, 4) be two points and C(x, y) be a point such that area of $\triangle ABC$ is 3 sq.units and $(x-1)(x-3) + (y-2)(y-4) = 0$. Then maximum number of positions of C, in the xy-plane is
- A) 2 B) 4 C) 8 D) no such C exist
73. If $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$ and $f(1) = 2$ and $a_r = f(r)$, for $r \in \mathbb{N}$ then the coordinates of a point on the parabola $y^2 = 8x$ whose focal distance is 4 may be
- A) (a_1, a_2) B) $(-a_1, -a_2)$ C) $(a_1, -2a_2)$ D) $(a_1, 2a_2)$
74. $\int_0^{1.5} x [x^2] dx =$ (where [.] denotes Greatest Integer Function)
- A) 0.5 B) 0.7 C) 0.75 D) 0.55

75. The area of the region bounded by the curves $y = x^2$ and $y = \frac{2}{1+x^2}$ is
- A) π B) $\pi - \frac{2}{3}$ C) $\pi - \frac{1}{3}$ D) $\pi + \frac{2}{3}$
76. If a, b, c, d are distinct integers in AP such that $d = a^2 + b^2 + c^2$ then the value of $a + b + c + d$ is
- A) 0 B) 1 C) 2 D) 3
77. If $\cos(\theta - \alpha) = a$ and $\sin(\theta - \beta) = b$ $\left(0 < \theta - \alpha, \theta - \beta < \frac{\pi}{2}, a, b \in [-1, 1]\right)$ then $\cos^2(\alpha - \beta) + 2ab\sin(\alpha - \beta)$ is equal to
- A) $a^2 - b^2$ B) $a^2 + b^2$ C) $2a^2b^2$ D) a^2b^2
78. $\lim_{x \rightarrow 0} \left(1^{\csc^2 x} + 2^{\csc^2 x} + \dots + n^{\csc^2 x}\right)^{\sin^2 x} = \underline{\hspace{2cm}}$
(where n is a positive integer and $n > 1$)
- A) 0 B) 1 C) n D) $n + 1$
79. The values of a, b and c which make the function $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & x < 0 \\ c, & x = 0 \\ \frac{\sqrt{x+bx^2} - \sqrt{x}}{bx^{3/2}} & x > 0 \end{cases}$ is continuous at $x = 0$, are
- A) $a = -\frac{3}{2}, c = \frac{1}{2}, b = 0$ B) $a = \frac{3}{2}, c = \frac{1}{2}, b \neq 0$
- C) $a = -\frac{3}{2}, c = \frac{1}{2}, b \neq 0$ D) $a = \frac{-3}{2}, c = \frac{-1}{2}, b = 0$
80. If $x = \sec \theta - \cos \theta$, $y = \sec^n \theta - \cos^n \theta$, then $\left(\frac{dy}{dx}\right)^2$ is (where $n \geq 2$ and $n \in N$)
- A) $\frac{n^2(y^2 + 4)}{x^2 + 4}$ B) $\frac{n^2(y^2 - 4)}{x^2}$ C) $n \frac{(y^2 - 4)}{(x^2 - 4)}$ D) $\left(\frac{ny}{x}\right)^2 - 4$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let $K_1 < K_2$ be the values of K (where $K \in \mathbb{R}$) for which the planes $Kx + 4y + z = 0$, $4x + Ky + 2z = 0$ and $2x + 2y + z = 0$ intersect in a straight line and $P = \begin{bmatrix} K_2 & K_1 \\ K_1 & K_2 \end{bmatrix}$. Then $\det(P) =$
82. If $3 \tan^{-1} \frac{1}{2 + \sqrt{3}} - \tan^{-1} \frac{1}{x} = \tan^{-1} \frac{1}{3}$ then $x =$
83. The number of distinct rational numbers of the form p/q , where $p, q \in \{1, 2, 3, 4, 5, 6\}$ is
84. One ticket is selected at random from 100 tickets numbered 00, 01, 02, , 99. Suppose X and Y are the sum and product of the digits found on the ticket respectively, then the value of $P(X = 7 / X = 0)$ is K , then $19K =$
85. The minimum value of $|x - 3| + |x - 2| + |x - 5|$ is
86. Let $\vec{OA} = \vec{a}$, $\vec{OB} = 10\vec{a} + 2\vec{b}$ and $\vec{OC} = \vec{b}$ where A and C are non collinear points. Let 'p' denote the area of quadrilateral $OABC$ and let 'q' denote the area of parallelogram with $\vec{OA}$ and $\vec{OC}$ as adjacent sides. If $p = kq$ then $k =$
87. The eccentricity of the conic defined by $\left| \sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2} \right| = 3$ is K , then $3K =$
88. If $\int \frac{(\sqrt{x})^5}{(\sqrt{x})^7 + x^6} dx = a \ln \left(\frac{x^K}{x^K + 1} \right) + C$ (C is constant), then the value of $2k$ is
89. Solution of the differential equation $\frac{dy}{dx} = \frac{xy}{1-x^2} + x.y^{1/2}$ is $\sqrt{y} = \frac{1}{5}(x^2 - 1) + c(x^2 - 1)^K$ (where $|x| > 1$), then $|4K| =$
90. If $f(x) = \frac{9^x}{9^x + 3}$ for all $x \in \mathbb{R}$ and the sum $f\left(\frac{1}{1996}\right) + f\left(\frac{2}{1996}\right) + \dots + f\left(\frac{1995}{1996}\right)$ is K then $\frac{2K}{5} =$